CATHOLIC JUNIOR COLLEGE
JC2 PRELIM EXAMINATION
Higher 2

Answers

CANDIDATE NAME	
CLASS	2T
INDEX NUMBER	

BIOLOGY

9744/01

Paper 1 Multiple Choice

28th AUGUST 2018
1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write and/or shade your name, NRIC / FIN number and HT group on the Answer Sheet in the spaces provided unless this has been done for you.

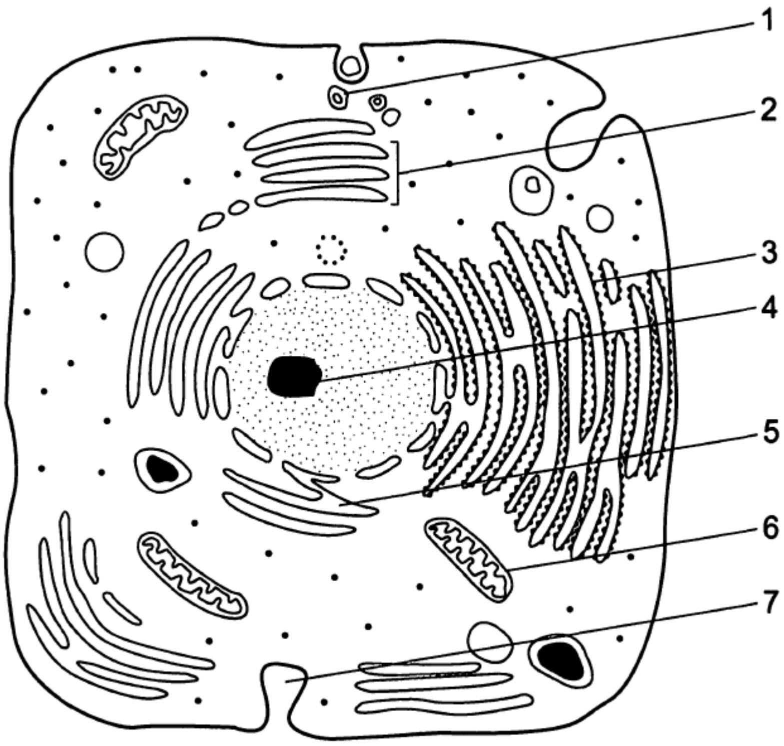
There are **thirty** questions on this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.
Choose the **one** you consider correct and record your choice in **soft 2B pencil** on the separate **OMR Answer Sheet.**

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
The use of an approved scientific calculator is expected, where appropriate.

1	B	2	C	3	B	4	C	5	A
6	D	7	A	8	B	9	B	10	C
11	C	12	D	13	B	14	D	15	C
16	C	17	A	18	B	19	D	20	D
21	B	22	B	23	A	24	C	25	C
26	B	27	B	28	B	29	D	30	C

1 The diagram shows a section of a generalised animal cell as seen under the electron microscope.

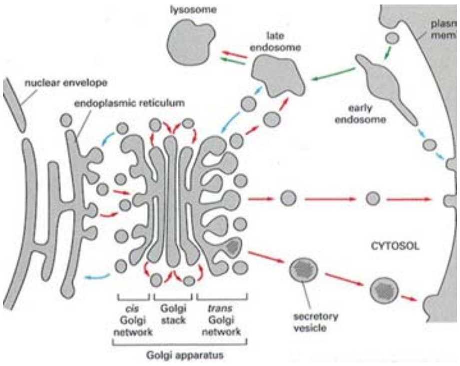


Where are the proteins and lipids synthesised and transported, packaged and secreted?

	synthesised and transport		packaged	secreted
	proteins	lipid	proteins and lipids	proteins and lipids
A	4	6	2	7
B	3	5	2	1
C	5	3	4	7
D	6	1	2	5

ANS B [L1] (H2 ALevel/2009/P1/Q1)
SC:
OR:

	Structure	Function	Macromolecule (location)
1	Secretary Vesicle (context given in question: proteins synthesised, transported, packaged & secreted) X not phagocytosis	Secretion	Proteins / lipids
2	GA ❼ Cis (nearer to RER/nucleus side) & Trans (nearer to CM)	Sorting and packaged	Proteins / lipids
3	ribosomes	Protein synthesis	Protein
4	nucleolus	rRNA synthesis	RNA
5	SER	Lipid synthesis	Lipids
6	Mitochondrion	ATP synthesis	-
7	Invagination made by CM	Bulk transport	Any substances



2 Which of the following statement describing the structures of proteins is **false**?

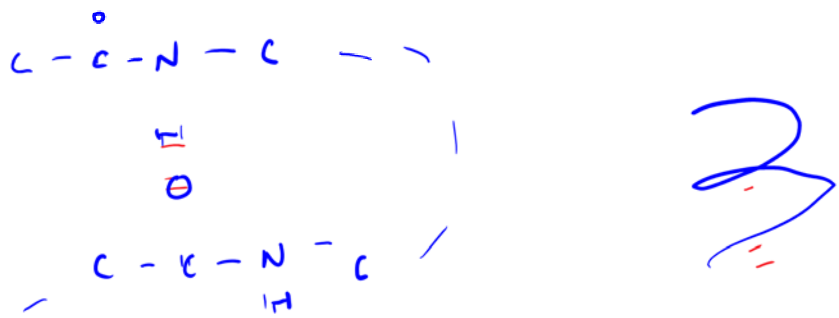
A The primary structure of a protein can determine if the polypeptide chain would form α-helix or β pleated sheet.

B Formation of intramolecular hydrogen bonds between peptide bonds is important to maintaining the shape of the secondary structures in proteins.

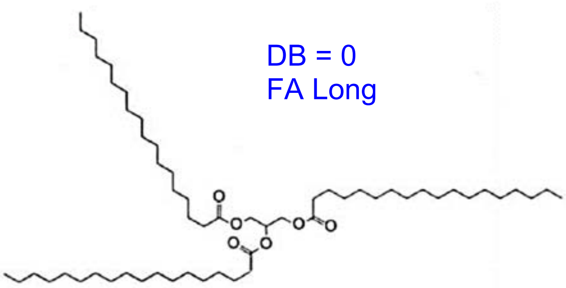
C The shape of a tertiary structure of protein can be determined by the types of interaction between the peptide bonds of amino acids far away from one another.

D A quaternary structure can be formed from non-covalent bonds between more than one polypeptide chains.

ANS C [L2] (H2 VJC/2015/P1/Q4 modified)
SC: structure of proteins **false**
OR:

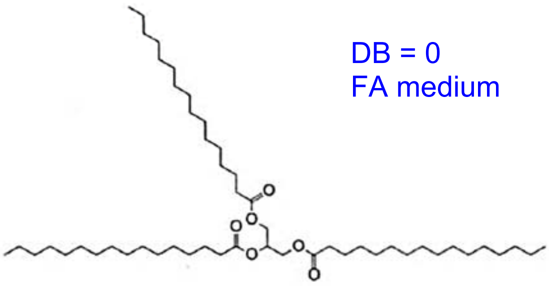
A	□ primary structure includes sequence and number of amino acids which will eventually determine if α-helix or β pleated sheet structures can be formed
B	□ 
C	□ interactions between the peptide bonds are part of what defines the secondary structure, and tertiary structure is defined by the bonds between the R-groups
D	□ types of bonds are not relevant to the definition of quaternary structure, only the number of polypeptides

3 The formulae and melting points of five triglycerides are shown in the diagram. Each triglyceride contains three identical fatty acids.



DB = 0
FA Long

tristearin 72 °C

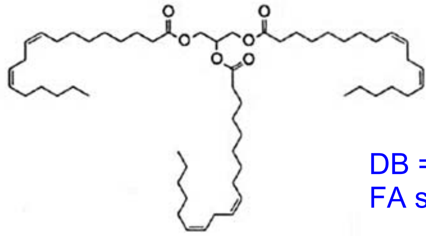


DB = 0
FA medium

tripalmitin 65.5 °C

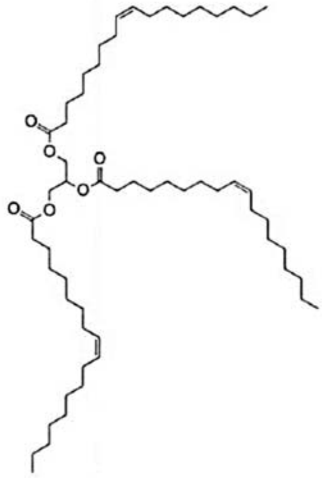
DB = 0
FA Shortest

trilaurin 46 °C



DB = 6
FA short

trilinolein –13 °C



DB = 3
FA medium

triolein –4 °C

Which two structural features of the molecules make the melting point higher?

	number of double bonds	length of fatty acid chains
A	fewer	shorter
B	fewer	longer
C	more	longer
D	more	shorter

ANS B [L1] (H1 Alevel/2012/P1/Q3)
SC: structural features **melting point higher**
OR: **more saturated (C-C)➐ higher melting points; less saturated (more C=C))➐ lower melting points**
longer chains ➐ more interactions ➐ higher melting points; shorter chains ➐ lower melting points

4 Which of the following statements about enzyme is false?

A both temperature and pH have similar negative effects on enzymes beyond the optimal ranges.

B both competitive and non-competitive inhibitors result in positive enzyme action though at only competitive inhibition can be overcome by increasing substrate concentration.

C enzymes lower the activation energy in reactions but have a threshold temperature for normal enzymatic reaction to take place.

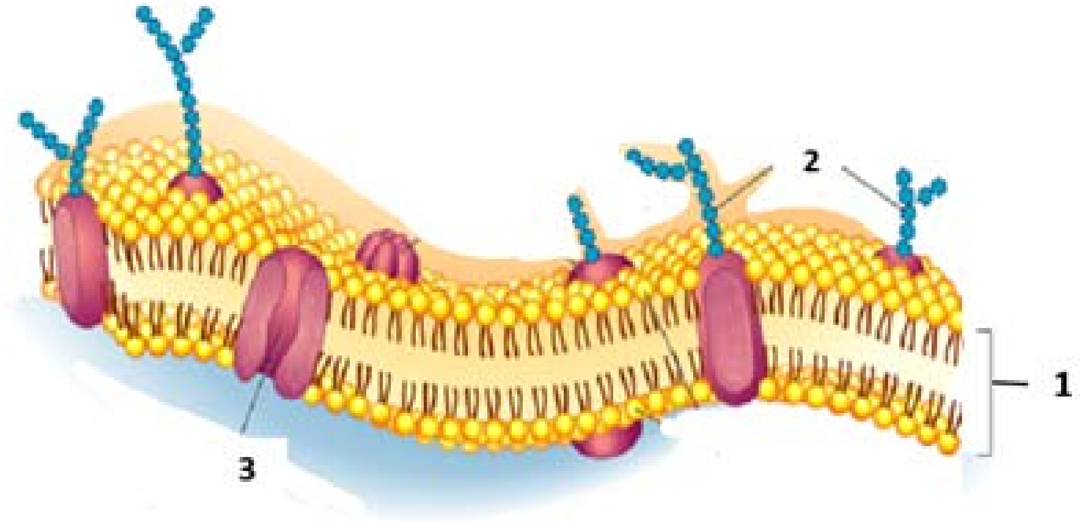
D there is a linear relationship between enzyme rate of reaction with both substrate concentration and enzyme concentration respectively, at low concentrations.

ANS B [L1] (Novel)
SC: Statements **False**
OR:
A T both temperature and pH have similar negative effects on enzymes beyond the optimal ranges.
B F both competitive and non-competitive inhibitors result in NEGATIVE enzyme action.

C **T** **enzymes lower the activation energy in reactions but have a threshold temperature for normal enzymatic reaction to take place.**

D **T** **there is a linear relationship between enzyme rate of reaction with both substrate concentration and enzyme concentration respectively, at low concentrations.**

5 The diagram below shows a representation of the plasma membrane.



Which of the following structures are correctly matched with their role(s)?

	cell-cell recognition	maintenance of glucose concentration	uptake of steroid hormones
A	**2** only	**1** and **3** only	**1** only
B	**2** only	**1** only	**1** and **3** only
C	**2** and **3** only	**1** only	**3** only
D	**3** only	**1** and **3** only	**1** only

ANS A **[L2]** (H2 RI/2016/P1/Q1)
SC: 1 = phospholipid bilayer; 2 = glycoprotein; 3 = transport protein
Correctly matched with roles
OR:

6 Blood transfusion laboratories around the world are hoping to produce large numbers of red blood cells (RBCs) from 'spare' human embryos produced during *in vitro* fertilisation procedures.

Embryonic stem cells are removed from an embryo and cultured in a growth medium that stimulates their differentiation into RBCs.

Which statement correctly describes this differentiation?

A Totipotent embryonic stem cells differentiate into pluripotent blood stem cells and then into RBCs.

B Totipotent embryonic stem cells differentiate into multipotent blood stem cells and then into RBCs.

C Pluripotent embryonic stem cells differentiate into totipotent blood stem cells and then into RBCs.

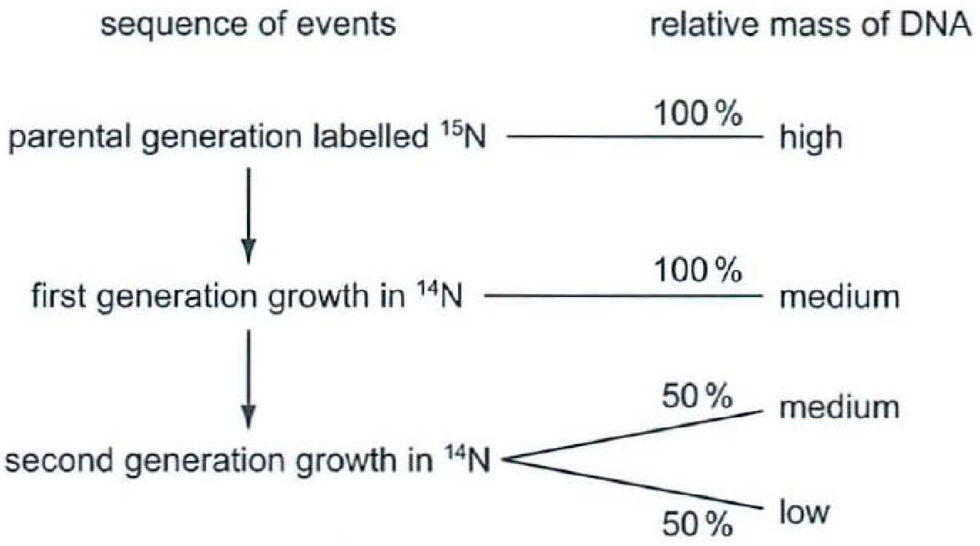
D Pluripotent embryonic stem cells differentiate into multipotent blood stem cells and then into RBCs.

ANS D [L1] (H1 Alevel/2013/P1/Q29 modified)
SC: statement **describe** **differentiation**
OR: **ES cells ➐ pluripotent; blood SC ➐ multipotent**

7 Meselson and Stahl investigated DNA replication by growing bacteria in culture containing heavy nitrogen, ^{15}N, until all the DNA was labelled.

These bacteria, the parental generation, were then transferred to a medium containing only light nitrogen, ^{14}N and allowed to replicate for two generations.

DNA was extracted from each generation of bacteria and its relative mass estimated. The flow diagram shows the results.



Which row explains the mass of the DNA in the second generation grown in ^{14}N?

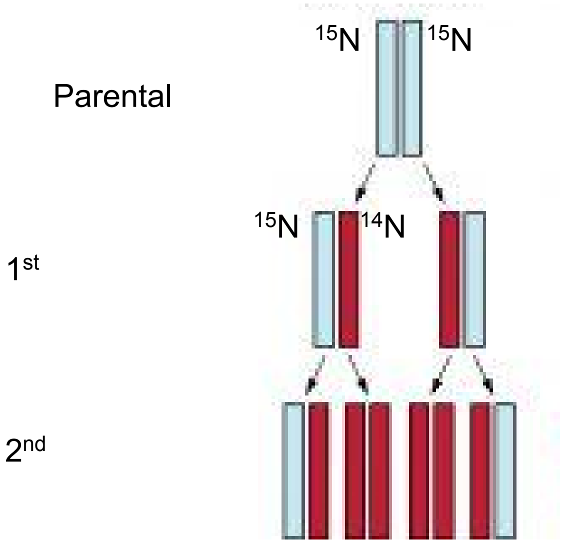
	50 % DNA molecules		50 % DNA molecules	
	strand 1	strand 2	strand 3	strand 4
A	only ^{15}N	only ^{14}N	only ^{14}N	only ^{14}N
B	only ^{15}N	only ^{15}N	only ^{14}N	only ^{14}N
C	only ^{15}N	only ^{15}N	only ^{14}N	only ^{15}N
D	only ^{15}N	half ^{14}N, half ^{15}N	only ^{14}N	half ^{14}N, half ^{15}N

ANS A [L2] (H1 Alevel/2012/P1/Q10)

SC: mass of DNA second generation grown in ^{14}N

OR: By strand 2nd generation ❼ 3 ratios of ^{14}N: 1 ratio of ^{15}N

	50 % DNA molecules		50 % DNA molecules	
	strand 1	strand 2	strand 3	strand 4
A	only ^{15}N √	only ^{14}N √	only ^{14}N √	only ^{14}N √
B	only ^{15}N √	only ^{15}N X	only ^{14}N √	only ^{14}N √
C	only ^{15}N √	only ^{15}N X	only ^{14}N √	only ^{15}N √
D	only ^{15}N √	half ^{14}N, half ^{15}N X	only ^{14}N √	half ^{14}N, half ^{15}N X



8 Exceptions to the universal genetic code are found in mammalian mitochondria, as shown in the table.

mRNA codon	in mammalian cytoplasm, codes for	in mammalian mitochondria, codes for
AGA	arginine	stop / termination
AGG	arginine	stop / termination
AUA	isoleucine	methionine
UGA	stop / termination	tryptophan

A short length of messenger RNA was synthesised with the following base sequence.

AUAAGAAGGUGA

How many peptide bonds would be formed by ribosomes translating this mRNA in mammalian cell cytoplasm and in mammalian mitochondria?

	mammalian cell cytoplasm	mammalian mitochondria
A	2	1
B	2	0
C	3	0
D	3	1

ANS B [L2] (H1 ALevel/2014/P1/Q10)
SC: peptide bonds of mRNA
OR: AUA-AGA-AGG-UGA
mammalian cell cytoplasm:

AUA AGA AGG UGA

Isoleucine arginine arginine stop

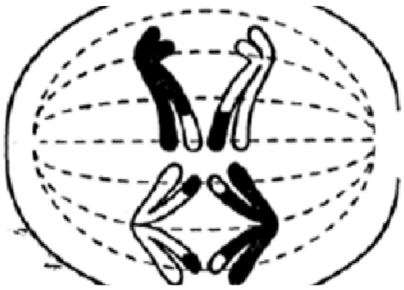
1 2 ➔ **2 peptide bonds**

mammalian mitochondria:

AUA AGA AGG UGA

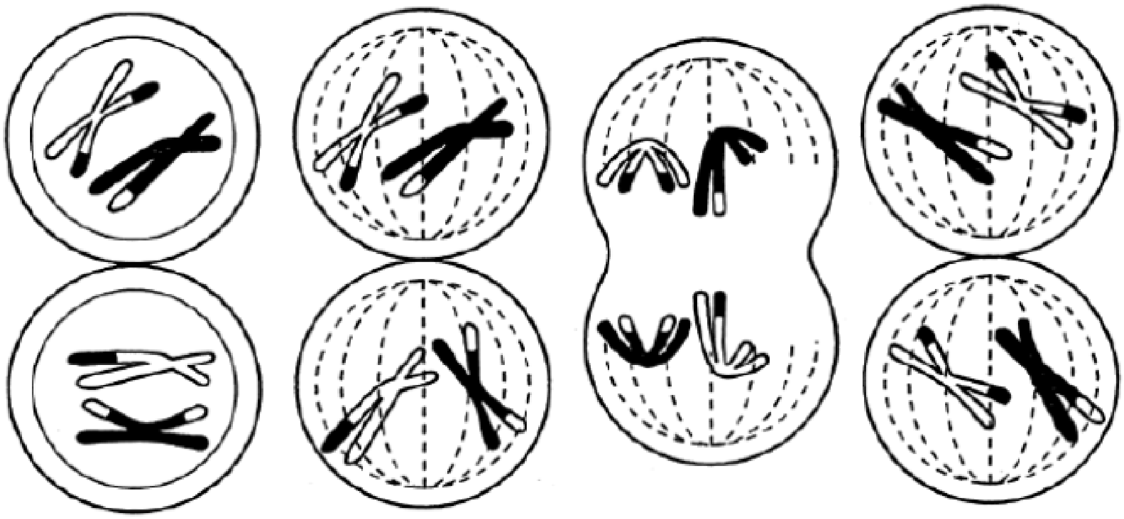
Methionine stop stop ❾ **0 peptide bonds**

9 The diagram shows anaphase I of meiosis.



Which diagram shows metaphase II as meiosis continues in this cell?

A **B** **C** **D**



ANS B [L2] (H2 SRJC/2016/P1/Q11)
SC: Anaphase I ❼ there is crossing over; metaphase II
OR: Only B and D are at metaphase II; D is showing the wrong crossed over chromosomes

10 Which of the statements about the p53 gene is false?

A It is used mainly in controlling uncontrolled cell division

B It is involved in regulating the cell cycle

C There is a need for both alleles to be present for suppression of cell division to take place.

D It is mutated or the effects of normal gene expression is not observed if cancer is found to be established.

ANS C [L1] (Novel)

SC:	**p53**	**False**	
OR:	**A**	**T**	**main function**
	B	**T**	**causes apoptosis in cancer cells**
	C	**F**	**only one allele needed**
	D	**T**	**not always mutated / non functional but can be overwhelmed.**

11 About 12,000 genes are expressed in both chick liver and oviduct. However, an estimated additional 5,000 genes are expressed only in liver, while an additional 3,000 genes are expressed only in oviduct.

1 The additional genes and their associated histones may have different methylation patterns in different tissues.

2 The number of genome copies is different in different somatic cells.

3 The number of transcriptional enhancer elements for the additional genes vary in different tissues.

4 A common set of genes are expressed for normal functions in liver and oviduct.

Which of the above could explain these observations?

A **1** only

B **2** and **3** only

C **1** and **4** only

D **3** and **4** only .

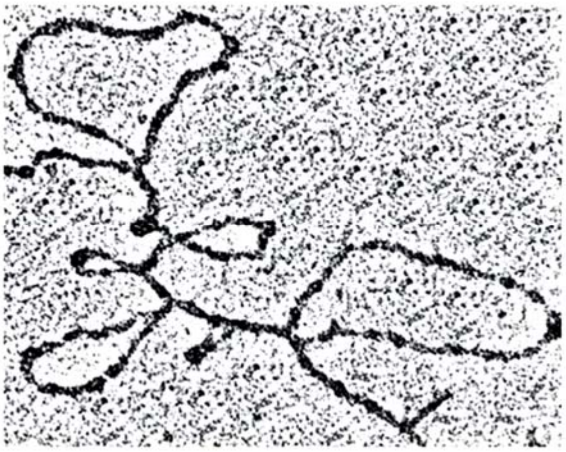
ANS C [L2] (H2 SAJC/2016/P1/Q17)

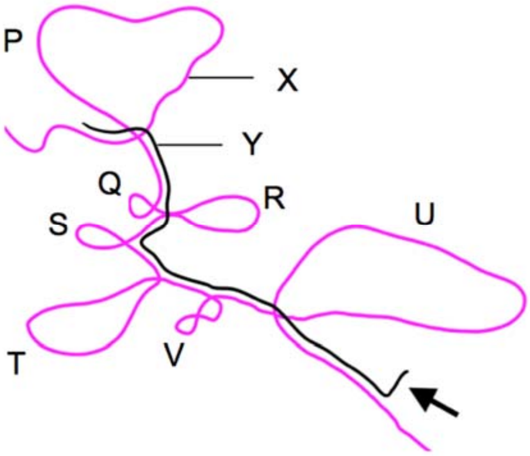
SC: 12000 for both; 5000 expressed only in liver vs 3000 only in oviduct

OR:

1	**T**; methylation on both DNA and histones will determine which genes are silenced
2	**F**; genome copy is the same for all cells
3	**F**; the transcriptional factor (proteins) may differ but the control elements are the same since it is part of the genome
4	**T**; some of these genes include “housekeeping genes” such as enzymes for respiration, DNA replication, transcription etc.

12 The ovalbumin gene from chicken was isolated and made single-stranded and then subsequently mixed together with its mature mRNA. The results were observed under an electron microscope. The electron micrograph and its corresponding diagrammatic representation show the binding of the mRNA to certain regions of the DNA.





Which of the following statement(s) can be concluded from the information above?

1 **X** is the template strand of DNA and **Y** is the mRNA strand transcribed from **X**.

2 **P, Q, R, S, T, U** and **V** correspond to the introns on the DNA that have been excised from the mRNA.

3 The arrow indicates the 3' end of the mRNA where the poly(A) tail was added during post-transcriptional modification.

4 The 3' end of the mRNA is free because there is no corresponding stretch on the template DNA where complementary base pairing can take place.

A **1** only

B **1** and **2** only

C **2**, **3** and **4** only

D All of the above

ANS D [L2] (H2 ACJC/2016/P1/Q17)
SC: gene isolated made single stranded and mixed with mature mRNA
OR:

1	**T**; the mRNA is transcribed from the template
2	**T**; the loops are not bound to the mRNA because their respective introns have been excised
3	**T**; the part of mRNA that is not bound to the DNA is likely to be the poly-A tail
4	**T**; as per 3 and that because poly-A tail (multiple nucleotides) is relatively much longer than 5' GTP cap (1 nucleotide long)

13 Which of the following statement(s) concerning *trp* operon is/are true?

1 A mutation of the gene coding for repressor protein will lead to the constitutive production of tryptophan.

2 There is one start and one stop codon in the mRNA of *trp* operon.

3 The repressor is inactive in the presence of excess tryptophan.

4 The mRNA codes for multiple polypeptides involved in the synthesis of tryptophan.

A **I** only

B **I** and **IV** only

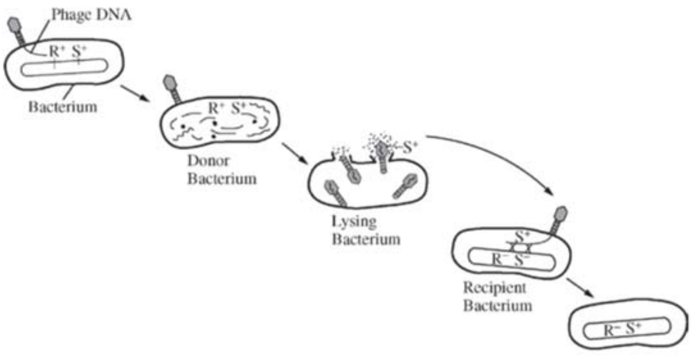
C **II** and **III** only

D **I, II** and **IV** only

ANS B [L1] (H2 RI/2016/P1/Q13 modified)
SC: true
OR:

1	**T**;
2	**F**; multiple genes in an operon
3	**F**; active in excess
4	**T**;

14 The diagram shows several steps how genetic variation in a population of bacteria can arise.



Which statement best describes this process?

A The recipient bacterium incorporates the transduced genetic material coding for phage proteins into its chromosome and synthesizes the corresponding proteins.

B Phage DNA undergoes recombination with DNA from the donor bacterium, and injects this new DNA into the recipient bacterium.

C The phage infection of the recipient bacterium and the introduction of DNA carried by the phage cause increased random point mutations of the bacterial chromosome.

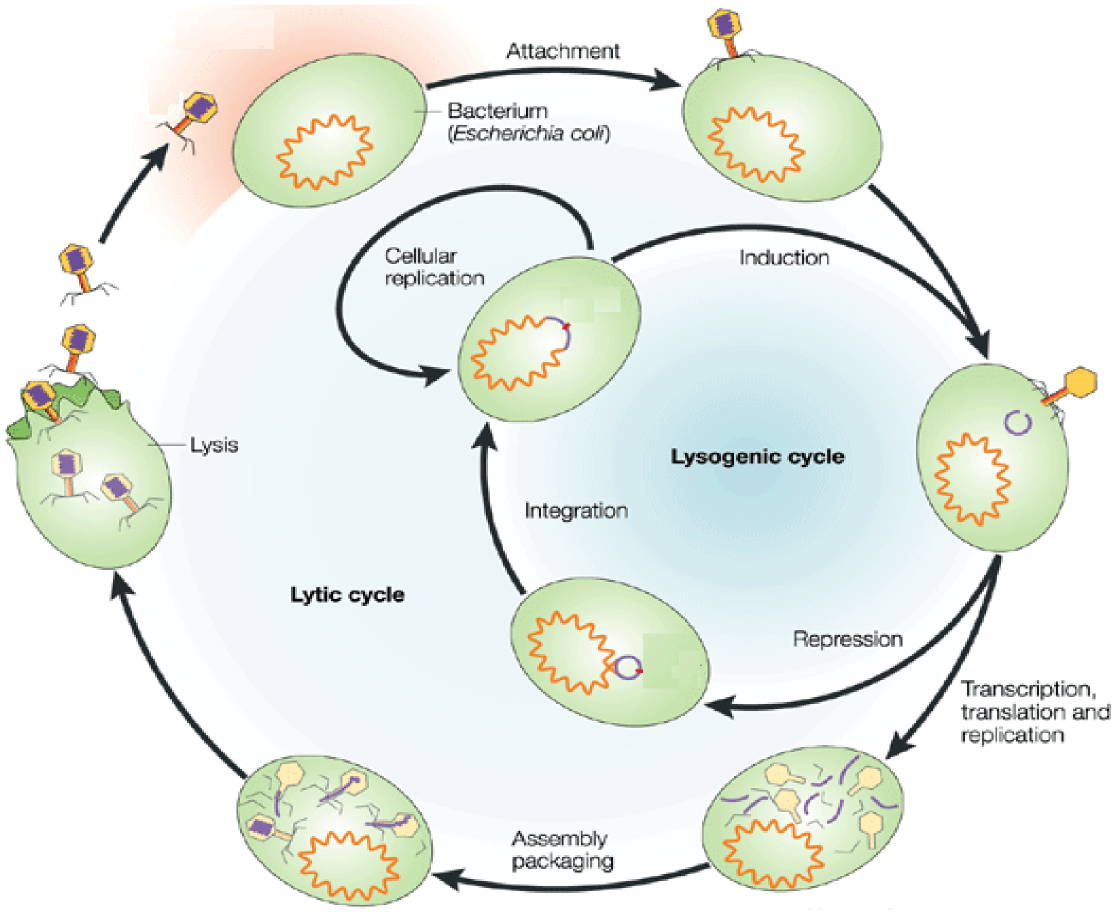
D DNA of the donor bacterium is lysed and then randomly packed into the phage, and this piece of DNA is then introduced to the recipient bacteria via the phage.

ANS D [L2] (H2 NJC/2016/P1/Q14 modified)
SC: best describes the process; **figure shows generalised transduction**
OR:

A	□ it is not a phage genome but the DNA from the host bacterium
B	□ there is no recombination with the host DNA because there are no compatible mechanism; recombination in recipient is possible because there is homology and compatibility
C	□ random mutation could potentially be caused by viral infection but is not shown in this diagram

D	▯

15 The diagram below shows the reproductive cycles of a phage infecting a host bacterium.



Some statements about the phages are listed as follows:

1. All types of phages are capable of undergoing lytic and lysogenic cycles.
2. A phage usually undergoes lysogenic cycle because a larger number of progeny phages can be produced rapidly as compared to the lytic cycle.
3. The release of lysozyme upon rupturing of lysosome leads to the osmotic lysis of host bacterium.
4. The phage gene codes for a repressor protein that prevents the expression of prophage.

Which of these statements about the reproductive cycles of a phage are **not** true?

A 1 and 2 only

B 3 and 4 only

C 1, 2 and 3 only

D 2, 3 and 4 only

ANS C [L2] (H2 YJC Prelim/2017/P1/Q9)
SC: statements **reproductive cycles of a phage** **NOT true**
OR:

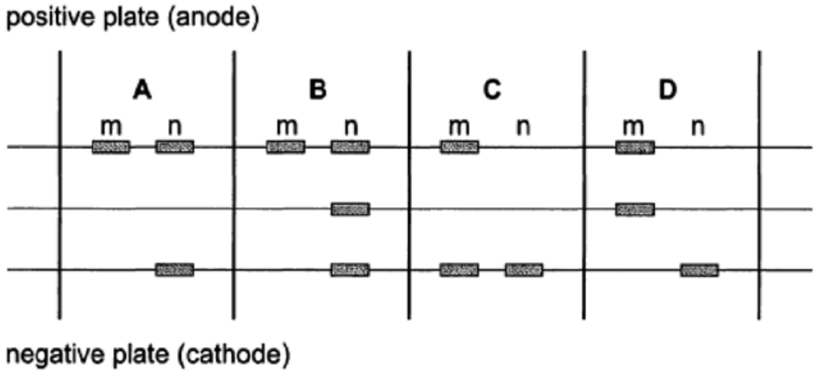
1	X not true. Not **all** types of phages can undergo both lytic & lysogenic. E.g. T-even phages only go thru lytic cycle
2	X not true. Lysogenic phages has their DNA integrated in host genome. They cannot produce large no. of phages **rapidly** at short time frame
3	X not true. Release of lysozyme upon rupture of cell membrane of bacteria host. Lysosome is an organelle and organelles are not found in bacterium / *E.coli*.
4	√ True.

16 There are two forms of the seeds of the garden pea, Pisum sativum, one with a wrinkled skin and the other with a smooth skin.

Plants with wrinkled seeds have a recessive mutation caused by the insertion of a transposable element into a gene.

Two heterozygous pea plants were crossed. Two of the offspring (m and n) had the DNA for this gene removed and separated by electrophoresis.

Which pattern of bands shows a wrinkled phenotype and a smooth phenotype?



ANS C [L3] (H2 GCE A/2009/P1/Q38)

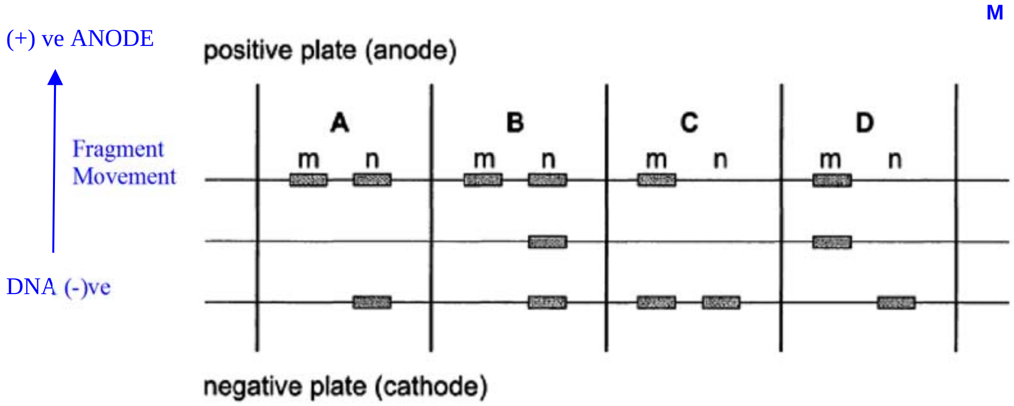
SC: Wrinkled Skin Recessive insertion of element **Smooth skin** **Ss x Ss**

OR: longer due to insertion **shorter**

SS **Short** **Smooth**

Ss **short long** **Smooth** **M**

ss **long** **wrinkled** **n**



Low scoring candidates found this question difficult as about one third chose option A and another third chose option D. Option A would indicates that the candidates thought that m was homozygous, but did not take into consideration that the wrinkled mutant DNA allele is longer and so would move less distance than the smooth non-mutant DNA. Alternatively they confused the direction of movement of the DNA. Option D might suggest some confusion with DNA evidence for semi-conservative replication as the heterozygote has an allele of intermediate mass.

17 Tortoiseshell coat colour in domestic cats is controlled by two genes where one is autosomal and the other is found on the X chromosome which determines sex.

B is dominant and is expressed as black coat colour.

b is recessive and is expressed as brown coat colour.

$\mathbf{X^G}$ is dominant and is expressed as ginger coat colour, and is expressed even in the presence of allele **B**

$\mathbf{X^g}$ is recessive and is expressed as absence of ginger colour.

During the development of female cats, one X chromosome in each cell is inactivated at random. Consequently, in female cats with the genotype $\mathbf{BBX^GX^g}$, skin cells may be in a patch of ginger fur or black fur depending on whether they have developed from a cell with an active $\mathbf{X^G}$ allele or $\mathbf{X^g}$ allele. Cats with this form of colouring are called tortoiseshell cats.

A black cat of genotype $\mathbf{BbX^gY}$ mated with a ginger cat of genotype $\mathbf{bbX^GX^g}$.

Assuming they have a large litter size of kittens, what proportion of their kittens would be expected to have the tortoiseshell colouring?

A 0.125

B 0.375

C 0.25

D 0.5

ANS A [L3] (H2 TPJC/2017/P1/Q19 modified)
SC:
OR:
X^G > B allele but X^g will show black colour.

♂ gametes / ♀ gametes	BX^g	bX^g	BY	bY
bX^G	BbX^GX^g Tortoiseshell	bbX^GX^g Ginger	BbX^GY Black	bbX^GY Brown
bX^g	BbX^gX^g Black	bbX^gX^g Brown	BbX^gY Black	bbX^gY Brown

18 The table below shows the results of an early investigation into the genetic control of phenotypic variation. The dry masses of 5493 bean seeds collected from many plants were classified into nine categories.

Mass of bean/mg	51-150	151-250	251-350	351-450	451-550	551-650	651-750	751-850	851-950
Number of beans	5	38	370	1676	2255	928	187	32	2

Which statement best describes these data and could account for the variation shown?

A The phenotypic variation is continuous and could be the result of two non-linked genes acting on their own.

B The phenotypic variation is continuous and could be the result of several non-linked genes acting on their own.

C The phenotypic variation is discontinuous and could be the result of two linked genes acting on their own.

D The phenotypic variation is discontinuous and could be the result of several linked genes acting on their own.

ANS B [L2] (H2 ACJC/2016/P1/Q23)
SC: phenotypic variation bellshaped curve data best describes
OR: additive effect of genes ➐ continuous variation; lesser genes involved = less bell shaped curve, data is narrower; more genes involved = data is more spread out

19 Flamingos are birds that live by lakes and only lay 1 egg per year. The feather colour of flamingos may vary from white to pink to red. To investigate the inheritance of feather colour, a scientist performed the following crosses over 3 years and recorded the feather colour of all the offspring when they were one-year-old. The diet of the offspring was also recorded.

cross	feather colour of parents	feather colour of all 3 one-year-old offspring	diet of offspring
1	white x white	white	aquatic plants
2	red x white	white	aquatic plants
3	white x white	pink	algae and crustaceans
4	red x white	pink	algae and crustaceans

Based on the above information, which two crosses best serve as evidence for the hypothesis that the feather colour of flamingos is influenced by their environment?

A **1** and **2**

B **1** and **3**

C **1** and **4**

D **2** and **4**

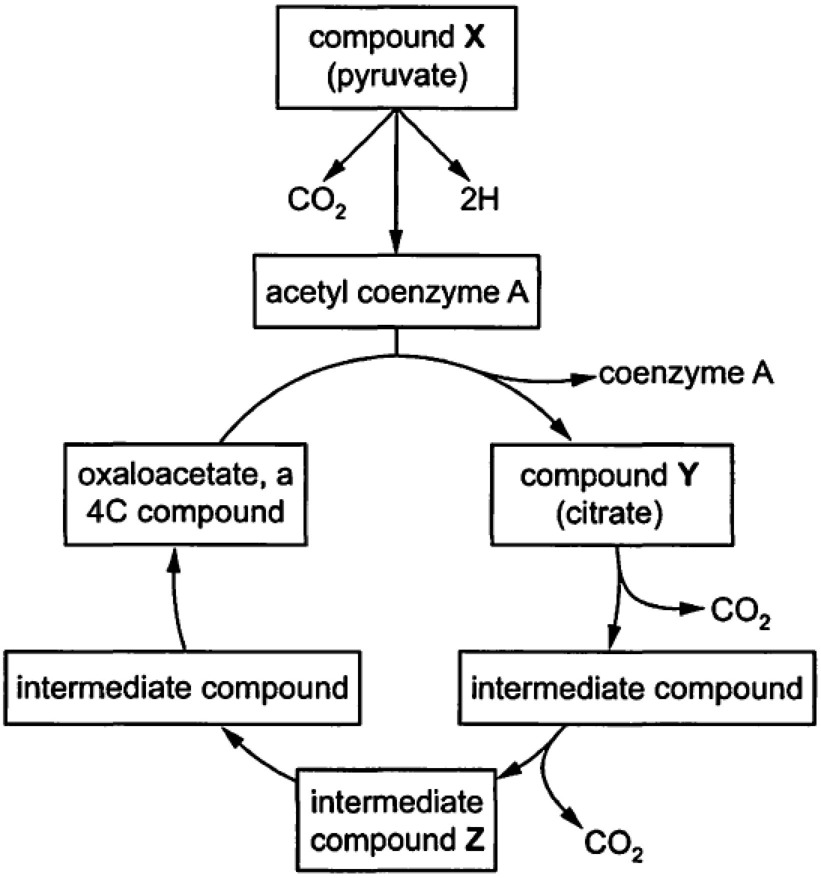
ANS D [L3] (H2 RI/2016/P1/Q20 modified)

SC: feather colour influenced by environment

OR: question requires student to be able to see which two crosses show that it is possible to be ONLY due to genetics and therefore does not require the influence of environment; as well as which cross is NOT mendelian genetically possible and therefore require the explanation of environmental influence.

1 and **2**	could be a case of complete dominance with white being the dominant allele; diet not changed
1 and **3**	could be a case of recessive allele for pink and white being dominant, with both parents heterozygous for cross 3
1 and **4**	cross 4 could be a case of just co-dominance
2 and **4**	cross 4 suggests co-dominance, but the same is not seen in cross 2. Therefore it must be due to the additional influence of diet

20 The diagram shows some of the reactions following glycolysis during aerobic respiration.



How many carbon atoms are in each molecule of compounds **X**, **Y** and **Z**?

	X	Y	Z
A	2	5	3
B	2	6	4
C	3	4	6
D	3	6	4

ANS D [L2] (H1 ALevel/2015/P1/Q18)
SC: Carbon atoms **X** **Y** **Z**
OR: Factual recall **3** **6** **4**

21 Where does ethanol formation occur in a yeast cell?

A cell vacuole

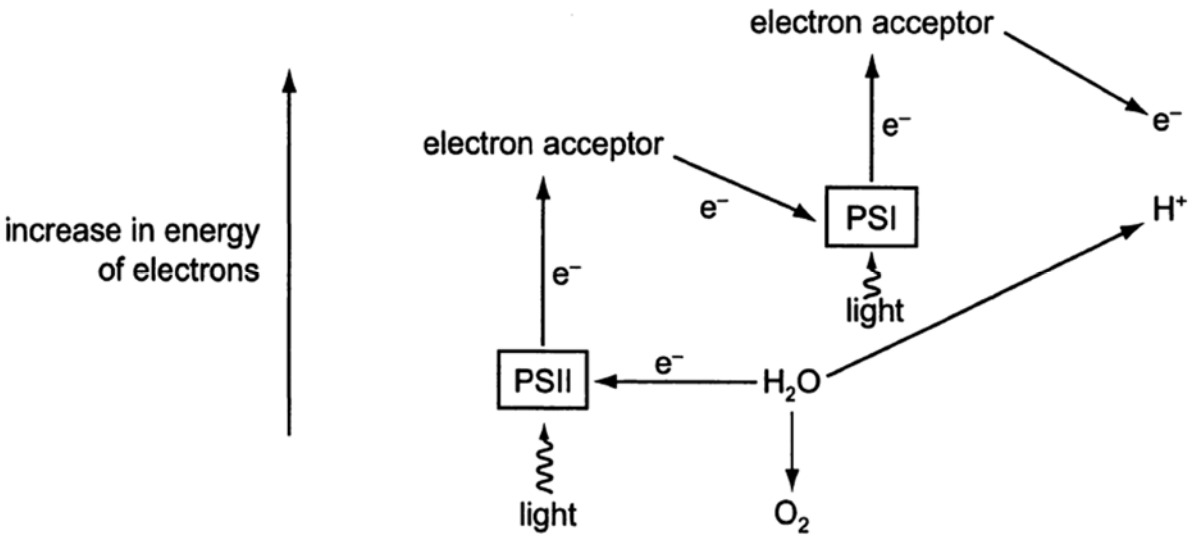
B cytoplasm

C Golgi apparatus

D mitochondrion

ANS B [L1] (H2 ALevel/2002/P1/Q24)
SC: ethanol formation
OR: in cytoplasm

22 The diagram shows the path taken by electrons and the formation of hydrogen ions in the light-dependent stages of photosynthesis.



What are the electrons and hydrogen ions used to produce?

A ATP from ADP

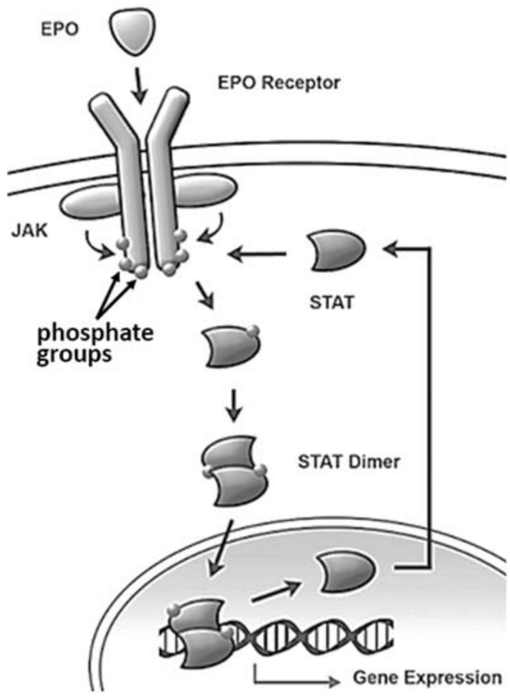
B ATP from ADP and reduced NADP from NADP

C glycerate 3-phosphate from glyceraldehyde 3-phosphate

D reduced NADP from NADP

ANS B [L1] (H2 Alevel/2009/P1/Q27)
SC: electrons **H + ions** **light dependent rxns**
OR: Products ➐ ATP and reduced NADPH

23 The diagram shows the JAK-STAT cell signalling pathway.



Which of the following statement is correct?

1 EPO phosphorylates JAK.

2 Phosphorylation of JAK causes EPO Receptors to undergo dimerisation.

3 Phosphorylation of STAT causes them to dimerisation.

4 Gene expression is terminated when phosphatases remove phosphate groups from STAT dimers.

5 Signal amplification occurs as JAK phosphorylates multiple tyrosine residues on the EPO receptor.

A 3 and 4 only

B 1, 3 and 4 only

C 3, 4 and 5 only

D 1, 2, 3 and 5 only

ANS A [L2] (H2 TJC/2017/P1/Q25 modified)
SC: correct
OR:

EPO phosphorylates JAK.	□ EPO binds to EPO receptors
Phosphorylation of JAK causes EPO Receptors to undergo dimerisation.	□ EPO receptors dimerises due to EPO
Phosphorylation of STAT causes them to dimerisation.	□ as seen in diagram
Gene expression is terminated when phosphatases remove phosphate groups from STAT dimers.	□ yes as when they are not phosphorylated they will dissociate and becomes not activated / functional
Signal amplification occurs as JAK phosphorylates multiple tyrosine residues on the EPO receptor.	□ there is no signal amplification here

24 Which of the following statements is true regarding the process of evolution?

A Rate of evolution is fastest if isolation and gene flow is greatest

B Mutations are advantageous and promote greater diversity in the gene pool.

C Genetic drift along with increase in allele frequency promotes evolution

D High reproductive potential and high degree of selection promote evolution.

ANS C [L2] (Novel)

SC: **True**
OR:
A] **Gene flow inhibits evolution as populations remain similar and less distinct.**
B] **Mutations are not always advantageous and may be deleterious.**
C] **Genetic drift allows slow but increased variation in the gene pool.**
D] **High degree of selection may not allow variation to take place soon enough for an advantageous gene to be present in the gene pool.**

25 A population of zooplankton is exposed to a small number of predatory fish that feed on the larger-sized adult zooplankton. Assuming no extinction takes place, which of the following predictions would most likely occur based on the principles of natural selection?

A The predatory fish will evolve smaller mouths so that they do not drive their prey to extinction.

B The small population of predatory fish will grow in number because of abundant food supply in the zooplankton.

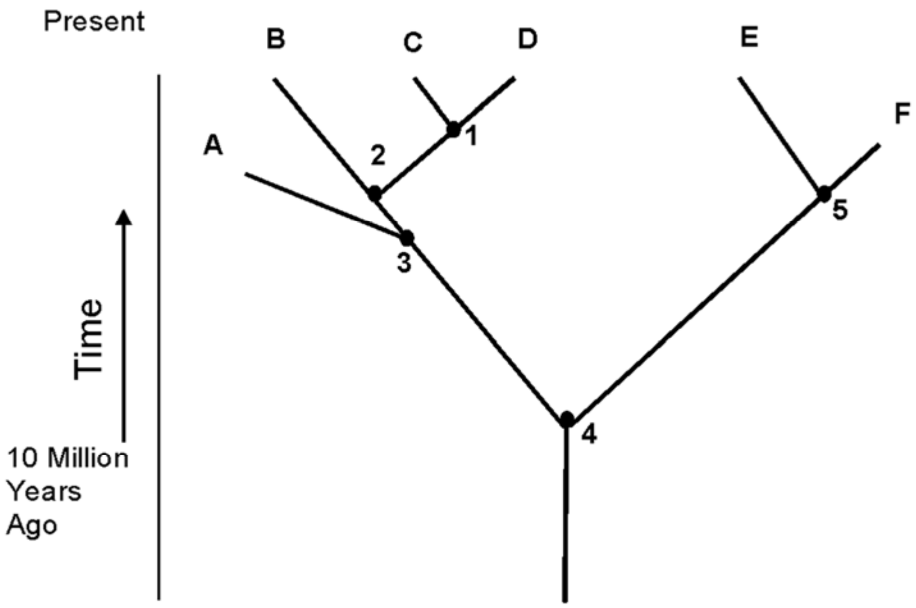
C Adult zooplankton will start to reach sexual maturity when they are still relatively small.

D More mutations will be observed in the zooplankton population to ensure a greater chance of survival.

ANS C [L2] (H1 RJC/2006/P2/Q29)

SC: **Zooplankton** **predator fish feed large ZP** **nat selection**
OR: **population** **selection pressure predation**
leave only small ZP
A] **Selection pressure does not serve in this manner**
B] **only Large ZP are fed upon**
C] **Mutation may lead to earlier maturity given no extinction takes place.**
D] **Not the purpose of Nat Selection. i.e. NS does not cause mutations**

26 The diagram below shows a phylogenetic tree.



Which of the following statements are true?

I. Organisms A, B, C, D, E and F are of different species, derived from a common ancestor.

II. Organisms A and B are more closely related than organisms E and F.

III. Organisms A and F are extinct.

IV. Organisms C share more homologous structures with B than with D.

A I, II and III only

B I and III only

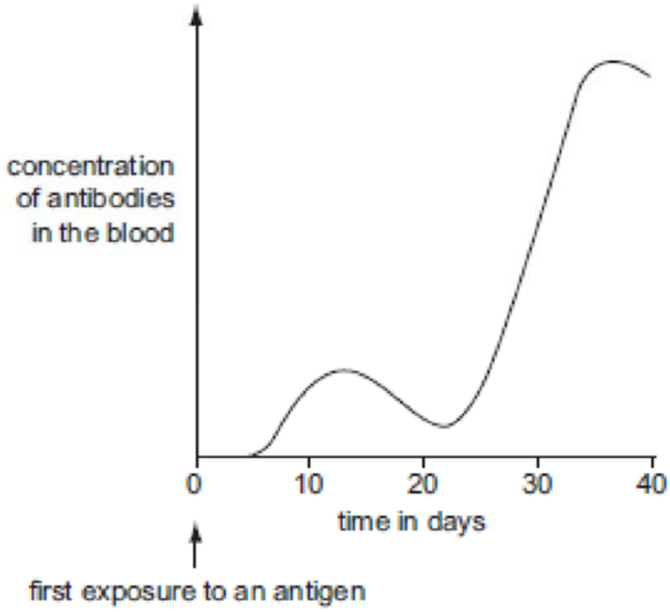
C II and IV only

D III and IV only

ANS B [L2] (H2 RJC/2016/P1/Q36)
SC: True
OR: **I TRUE – given common ancestor is 4**
II FALSE 3 is Common Ancestor B descendent; E and F descendants more closely related. A&B are both descendants, but 2 evolutionary events for 3 to B have occurred while only one is seen for E to F. Moreover A&B diverged earlier than E&F which is a more recent common ancestor.
III TRUE
IV FALSE – All 3 share the same Homologous structures, C & D share more compared to B

27 The graph shows the amount of antibody produced in response to an antigen.

Fig. 27.1



From the graph, which statement is correct?

A It takes 25 days to achieve active immunity.

B Memory cells for this antigen are present in the body within 20 days.

C T helper cells are activated on day 12.

D The second exposure to the antigen occurred on day 25.

ANS B [L2] (9700/s07.Q37)
SC: **Correct**
OR: **A] Active immunity achieved at 12Days**
B] Only with memory cells will the second response be faster and more significant.
C] Antibodies are already created at D5-6 meaning T helper had already been active before that.
d] Second response was at Day 22/23 already showing 2nd increase in antibodies.

28 Which future development in vaccine production is most important in the fight to eradicate measles in developing countries?

A a combined vaccine to combat it and other diseases

B a single vaccine, without the need for boosters

C a vaccine containing only live measles viruses

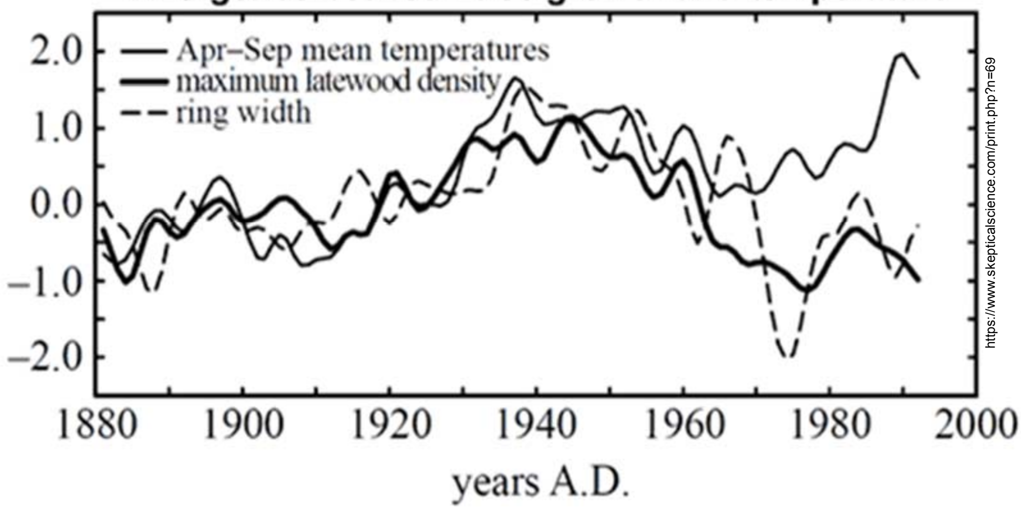
D a vaccine produced by genetic engineering techniques

ANS B [L2 (9700/s08.Q37)
SC: **Future development vaccine imp eradicate measles developing countries**
OR: **A] combined vaccine may be harder to administer in dev countries, may cost more if other diseases are included**
B] Targetted, without boosters would give the ability for sustained coverage.
C] All strains of measles are virulent and not appropriate for live introduction.
D] Expensive and non productive as mutations may render vaccine ineffective.

29 Tree growth is sensitive to temperature. Consequently, tree-ring width and tree-ring density, both indicators of tree growth, serve as useful proxies for temperature. By measuring tree growth in ancient trees, scientists can reconstruct temperature records going back over 1000 years. Comparisons with direct temperature measurements back to 1880 show a high correlation with tree growth. However, in high latitude sites, the correlation breaks down toward the end of the 19th century.

Divergence between tree growth and temperature

Fig. 29.1



Which of the following best explains the breakdown in correlation seen in Fig. 29.1?

A Biotic factors may have affected tree growth in both medium and high latitude sites

B Global warming from 1985 onwards has affected tree growth significantly.

C Latewood density below 0.0 A.U. is not a good proxy for temperature.

D There are other non-natural factors affecting tree growth from the 1960s onward.

ANS D [L3 (Novel)

SC: **Best explains** **breakdown correlation**
OR: **1880-1960** **1960 to 2000**
close correlation **temp rises independently**
Latewood density decline [threshold?]
industrial revolution 1960s

A] Only high latitudes mentioned.
B] decline correlation was before 1985, approx. 1960s
C] Latewood density has been lower than 0.0 A.U. before but not affected correlation.
D] Anthropomorphic climate change affecting latewood trees.

30 Over the past century, dengue has had a positive trend in the widening of its range globally. Which statement/s about the wider spread of dengue is true?

I. Vector adaptations to lower temperature ranges allow wider latitudinal ranges

II. Wide temperature tolerances in vectors allow an increase in metabolism resulting in a shorter developmental stage.

III. Increased human population globally has allowed greater foraging advantage.

IV. increased density in human populations has allowed survival of the virus in regions beyond its thermal limitations.

A I only

B I and III only

C IV only

D II and IV only

ANS C [L2] (Novel)

SC: dengue range positive trend True

OR: I. □ Vectors have not adapted to lower temperatures, too short a time for an advantageous allele to arise.

II. □ Vectors do not have a wide temperature tolerance.

III. □/□ Humans are not fed upon for food, rather for reproductive purposes.

IV. □ cities are warmer and the increased temperature is advantageous to the Vector even in an intolerable colder region.

END OF PAPER